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Batch: 15

Experiment-49

Algorithm

Algorithm: Pivot Integer Using Pointer

1. Start.
2. Declare integer variable n.
3. Read the value of n.
4. Check whether n is positive.
5. If n <= 0, display **"Please enter a positive integer"** and stop.
6. Pass the address of n to the function pivot_integer() using &n.
7. In pivot_integer(), receive the address using pointer *n.
8. Repeat for x from 1 to *n:
    - Set leftSum = 0.
    - Set rightSum = 0.
    - Calculate the sum of integers from 1 to x and store it in leftSum.
    - Calculate the sum of integers from x to *n and store it in rightSum.
    - Compare leftSum and rightSum.
    - If they are equal:
        - Display x as the pivot integer.
        - Return from the function.
9. If no value of x satisfies the condition, display -1.
10. Stop.

Test Case Table

Test Case	Input n	Expected Output	Reason
1	8	Pivot integer: 6	1+2+3+4+5+6 = 6+7+8 = 21
2	1	Pivot integer: 1	1 = 1
3	4	-1	No pivot integer exists
4	49	Pivot integer: 35	Sum from 1 to 35 = sum from 35 to 49
5	2	-1	No pivot integer exists
6	0	Please enter a positive integer.	Invalid input because n must be positive

C Program

```
#include <stdio.h>
```

```
void pivot_integer(int *n)
{
   int x, i;
   int leftSum, rightSum;

   for (x = 1; x <= *n; x++)
   {
      leftSum = 0;
      rightSum = 0;

      // Sum from 1 to x
      for (i = 1; i <= x; i++)
      {
         leftSum = leftSum + i;
      }

      // Sum from x to n
      for (i = x; i <= *n; i++)
      {
         rightSum = rightSum + i;
      }

      if (leftSum == rightSum)
      {
         printf("Pivot integer: %d\n", x);
         return;
      }
   }

   printf("-1\n");
}

int main()
{
   int n;

   printf("Enter a positive integer n: ");
   scanf("%d", &n);

   if (n <= 0)
   {
      printf("Please enter a positive integer.\n");
      return 0;
   }

   // Passing address of n
   pivot_integer(&n);

   return 0;
```

}


┌──(kali㉿kali)-[~]
└─$ vi hdoc49.c
┌──(kali㉿kali)-[~]
└─$ gcc hdoc49.c -o hdoc49
┌──(kali㉿kali)-[~]
└─$ ./hdoc49
Enter a positive integer n: 8
Pivot integer: 6
┌──(kali㉿kali)-[~]
└─$ gcc hdoc49.c -o hdoc49
┌──(kali㉿kali)-[~]
└─$ ./hdoc49
Enter a positive integer n: 4
-1
┌──(kali㉿kali)-[~]
└─$ gcc hdoc49.c -o hdoc49
┌──(kali㉿kali)-[~]
└─$ ./hdoc49
Enter a positive integer n: 0
Please enter a positive integer.